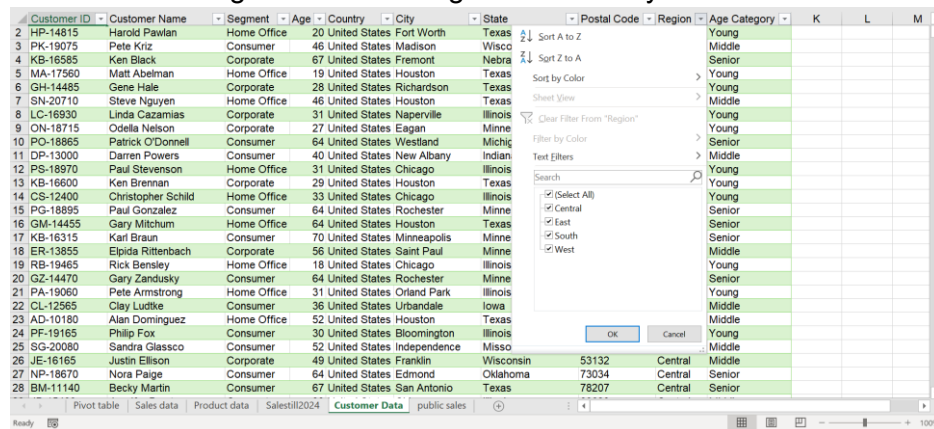


Project – Power Query, Power Pivot, and DAX

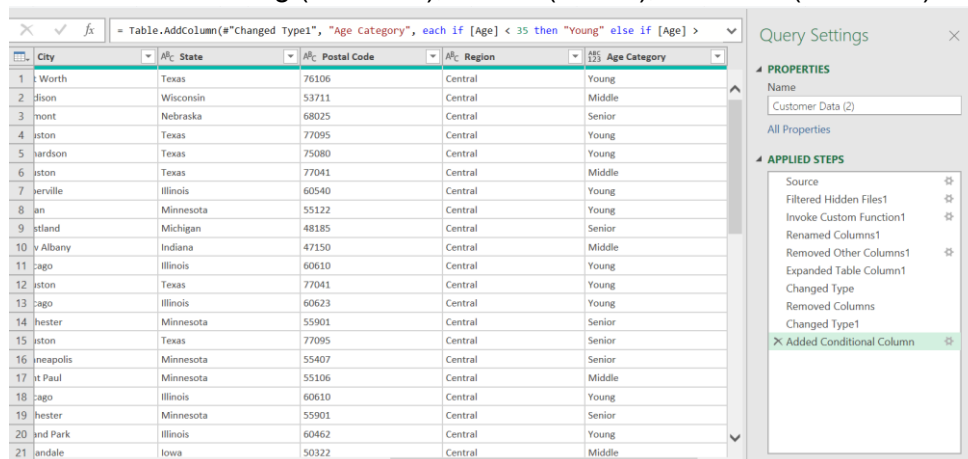
In this project, I explored the complete process of data preparation, modeling, and analysis using Power Query, Power Pivot, and DAX (Data Analysis Expressions). The objective was to consolidate and analyze customer, sales, and product data that existed in different formats and across multiple sources. Below are the key steps I followed:

1. I began by using the Get Data feature in Excel to import customer data maintained in separate CSV files by different regional teams. Rather than opening and merging each file manually, I used the "Get Data > From Folder" option to combine all customer files into a single dataset through Power Query.



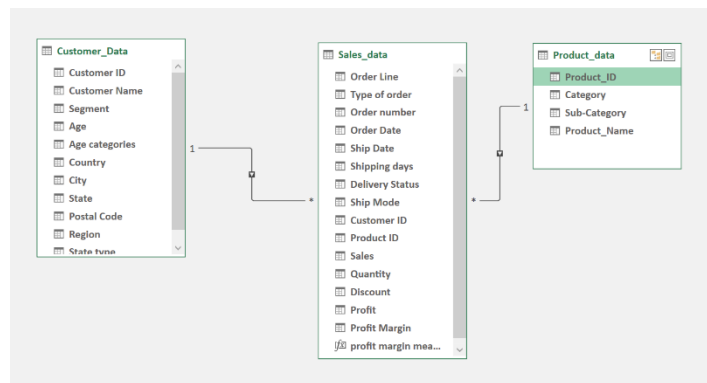
Customer ID	Customer Name	Segment	Age	Country	City	State	Postal Code	Region	Age Category	K	L	M
HP-14815	Harold Pawlan	Home Office	20	United States	Fort Worth	Texas			Young			
PK-19075	Pete Kriz	Consumer	46	United States	Madison	Wisconsin			Middle			
KB-16585	Ken Black	Corporate	67	United States	Fremont	Nebraska			Senior			
MA-17560	Matt Abelman	Home Office	19	United States	Houston	Texas			Young			
GH-14485	Gene Hale	Corporate	28	United States	Richardson	Texas			Young			
SN-20710	Steve Nguyen	Home Office	46	United States	Houston	Texas			Middle			
LC-16930	Linda Cazamas	Corporate	31	United States	Naperville	Illinois			Young			
ON-18715	Odella Nelson	Corporate	27	United States	Eagan	Minnesota			Young			
PO-18865	Patrick O'Donnell	Consumer	64	United States	Westland	Michigan			Senior			
DP-13000	Darren Powers	Consumer	40	United States	New Albany	Indiana			Middle			
PS-18970	Paul Stevenson	Home Office	31	United States	Chicago	Illinois			Young			
KB-16600	Ken Brennan	Corporate	29	United States	Houston	Texas			Young			
CS-12400	Christopher Schild	Home Office	33	United States	Chicago	Illinois			Young			
PG-18995	Paul Gonzalez	Consumer	64	United States	Rochester	Minnesota			Senior			
GM-14455	Gary Mitchum	Home Office	64	United States	Houston	Texas			Senior			
KB-16315	Karl Braun	Consumer	70	United States	Minneapolis	Minnesota			Senior			
ER-13855	Elpidia Rittenbach	Corporate	56	United States	Saint Paul	Minnesota			Middle			
RB-19465	Rick Bensley	Home Office	18	United States	Chicago	Illinois			Young			
GZ-14470	Gary Zandusky	Consumer	64	United States	Rochester	Minnesota			Senior			
PA-19060	Pete Armstrong	Home Office	31	United States	Orland Park	Illinois			Young			
CL-12565	Clay Ludtke	Consumer	36	United States	Urbana	Iowa			Middle			
AD-10180	Alan Dominguez	Home Office	52	United States	Houston	Texas			Middle			
PF-19165	Philip Fox	Consumer	30	United States	Bloomington	Illinois			Young			
SG-20080	Sandra Glassco	Consumer	52	United States	Independence	Missouri			Middle			
JE-16165	Justin Ellison	Corporate	49	United States	Franklin	Wisconsin	53132	Central	Middle			
NP-18670	Nora Paige	Consumer	64	United States	Edmond	Oklahoma	73034	Central	Senior			
BM-11140	Becky Martin	Consumer	67	United States	San Antonio	Texas	78207	Central	Senior			

2. In Power Query Editor, I performed data transformation tasks including:
 - a. Removing unnecessary columns such as the source filename.
 - b. Converting columns like postal codes from number to text, as they are identifiers rather than numerical values.
 - c. Creating a new "Age Category" column using conditional logic: classifying customers as Young (below 35), Middle (35–60), or Senior (above 60).



City	State	Postal Code	Region	Age Category
Worth	Texas	76106	Central	Young
Madison	Wisconsin	53711	Central	Middle
Fremont	Nebraska	68025	Central	Senior
Houston	Texas	77095	Central	Young
Richardson	Texas	75080	Central	Young
Houston	Texas	77041	Central	Middle
Naperville	Illinois	60540	Central	Young
Eagan	Minnesota	55122	Central	Young
Westland	Michigan	48185	Central	Senior
New Albany	Indiana	47150	Central	Middle
Chicago	Illinois	60610	Central	Young
Houston	Texas	77041	Central	Young
Chicago	Illinois	60623	Central	Young
Houston	Minnesota	55901	Central	Senior
Minneapolis	Texas	77095	Central	Senior
Minneapolis	Minnesota	55407	Central	Senior
Saint Paul	Minnesota	55106	Central	Middle
Chicago	Illinois	60610	Central	Young
Houston	Minnesota	55901	Central	Senior
Urbana	Illinois	60462	Central	Young
Urbana	Iowa	50322	Central	Middle

3. Once the data was cleaned and transformed, I loaded it into Excel and created a data model using Power Pivot. I enabled the Power Pivot add-in and added all relevant tables - Customer, Sales, and Product - to the data model. In the Power Pivot diagram view, I established relationships between the tables using common keys (e.g., Customer ID and Product ID). This allowed for relational analysis across multiple datasets.



4. After establishing relationships, I created Pivot Tables using fields from different tables. For example, I generated a pivot table showing Age Category-wise Sales by dragging the "Age Category" field from the customer table and sales values from the sales table.

A	B	C	D	E	F	G	H	I	J	K
Row Labels	profit margin measure									
Middle aged	0.109484381									
Senior	0.130939107									
Young	0.138840877									
Grand Total	0.124748641									

5. To perform more advanced calculations, I used DAX to create both calculated columns and explicit measures:
 - a. I created a calculated column named "State Type" to classify states like California, New York, and Texas as "Top" states, and all others as "Others".

[State t...]											
=if((Customer_Data[State]="Texas" Customer_Data[State]="New York" Customer_Data[State]="California"),"Top","Others")											
Custo...	Customer N...	Segm...	A...	Age catego...	Cou...	C...	St...	Postal C...	Regi...	State type	Add...
1	HP-14815	Harold Pawlan	Home Of...	20	Young	United ...	Fort...	Texas	76106	Central	Top
2	PK-19075	Pete Kriz	Consumer	46	Middle aged	United ...	Ma...	Wisc...	53711	Central	Others
3	KB-16585	Ken Black	Corporate	67	Senior	United ...	Fre...	Nebr...	68025	Central	Others
4	MA-17560	Matt Abelman	Home Of...	19	Young	United ...	Hou...	Texas	77095	Central	Top
5	GH-14485	Gene Hale	Corporate	28	Young	United ...	Rich...	Texas	75080	Central	Top
6	SN-20710	Steve Nguyen	Home Of...	46	Middle aged	United ...	Hou...	Texas	77041	Central	Top
7	LC-16930	Linda Cazamias	Corporate	31	Middle aged	United ...	Nap...	Illinois	60540	Central	Others
8	ON-18715	Odella Nelson	Corporate	27	Young	United ...	Eagan	Minn...	55122	Central	Others
9	PO-18865	Patrick O'Donnell	Consumer	64	Senior	United ...	Wes...	Michi...	48185	Central	Others
10	DP-13000	Darren Powers	Consumer	40	Middle aged	United ...	Ne...	Indiana	47150	Central	Others
11	PS-18970	Paul Stevenson	Home Of...	31	Middle aged	United ...	Chic...	Illinois	60610	Central	Others
12	KB-16600	Ken Brennan	Corporate	29	Young	United ...	Hou...	Texas	77041	Central	Top
13	CS-12400	Christopher Sch...	Home Of...	33	Middle aged	United ...	Chic...	Illinois	60623	Central	Others
14	PG-18895	Paul Gonzalez	Consumer	64	Senior	United ...	Roc...	Minn...	55901	Central	Others
15	GM-14455	Gary Mitchum	Home Of...	64	Senior	United ...	Hou...	Texas	77095	Central	Top
16	KB-16315	Karl Braun	Consumer	70	Senior	United ...	Min...	Minn...	55407	Central	Others

- b. I also created explicit measure - Profit Margin. This measure allowed for consistent, reusable, and context-aware calculation in Pivot Tables.

[Order ...]															
=profit margin measure:=[Sum of Profit]/[Sum of Sales]															
O...	Typ...	Order nu...	Order ...	Ship D...	Shipping d...	Delivery St...	Ship M...	Custo...	Pro...	Sa...	Qua...	Disco...	Pr...	Profit M...	
1	Offline	103800	6/26/2021 ...	6/30/202...		4	Late	Standard C...	DP-13000	OFF-PA-10...	16.45	2	20	5.5512	33.7458966...
2	Offline	112326	6/27/2021 ...	7/1/2021 ...		4	Late	Standard C...	PO-19195	OFF-LA-10...	11.78	3	20	4.2717	36.2623089...
3	Offline	112326	6/27/2021 ...	7/1/2021 ...		4	Late	Standard C...	PO-19195	OFF-ST-10...	272.74	3	20	-64.7...	-23.7496516...
4	Offline	112326	6/27/2021 ...	7/1/2021 ...		4	Late	Standard C...	PO-19195	OFF-BI-10...	3.54	2	80	-5.487	-155
5	Offline	141817	6/28/2021 ...	7/5/2021 ...		7	Late	Standard C...	MB-18085	OFF-AR-10...	19.54	3	20	4.884	24.9948822...
6	Offline	130813	6/29/2021 ...	7/1/2021 ...		2	On time	Second Class	LS-17230	OFF-PA-10...	19.44	3	0	9.3312	48
7	Offline	106054	6/29/2021 ...	6/30/202...		1	On time	First Class	JO-15145	OFF-AR-10...	12.78	3	0	5.2398	41
8	Offline	167199	6/29/2021 ...	7/3/2021 ...		4	Late	Standard C...	ME-17320	FUR-CH-1...	2573.82	9	0	746.4...	29
9	Offline	167199	6/29/2021 ...	7/3/2021 ...		4	Late	Standard C...	ME-17320	OFF-BI-10...	609.98	2	0	274.4...	45
10	Offline	167199	6/29/2021 ...	7/3/2021 ...		4	Late	Standard C...	ME-17320	OFF-AR-10...	5.48	2	0	1.4796	27
11	Offline	167199	6/29/2021 ...	7/3/2021 ...		4	Late	Standard C...	ME-17320	TEC-PH-10...	391.98	2	0	113.6...	29
12	Offline	167199	6/29/2021 ...	7/3/2021 ...		4	Late	Standard C...	ME-17320	TEC-PH-10...	755.96	4	0	204.1...	27
13	Offline	167199	6/29/2021 ...	7/3/2021 ...		4	Late	Standard C...	ME-17320	OFF-FA-10...	31.12	4	0	0.3112	1
14	Offline	167199	6/29/2021 ...	7/3/2021 ...		4	Late	Standard C...	ME-17320	OFF-PA-10...	6.54	1	0	3.0084	46
15	Offline	105417	6/30/2021 ...	7/5/2021 ...		5	Late	Standard C...	V5-21820	FUR-FU-10...	76.73	3	60	-53.7...	-69.9981754...
16	Offline	105417	6/30/2021 ...	7/5/2021 ...		5	Late	Standard C...	V5-21820	OFF-BI-10...	10.43	7	80	-18.2...	-175
prof...															